

Part A · Multiple Choice

2 points each

1. **a) the process by which compacted sediments become cemented together into solid rock**
p. 135 — it happens at depths of hundreds to thousands of meters and involves two steps: compaction from burial, then cementation as minerals crystallize in the pores between clasts.
2. **d) quartz is more resistant to weathering than any other common mineral**
p. 135 — by contrast, most clasts smaller than sand are clay minerals, and most clasts larger than sand are actual fragments of rock.
3. **a) conglomerate**
p. 140 — breccia has angular clasts. Conglomerates form in high-energy settings such as fast-flowing rivers where particles get rounded; breccias form where particles were not carried far, such as alluvial fans and talus slopes.
4. **b) Limestone**
p. 141 — almost all limestone forms in the oceans, mostly on shallow continental shelves in tropical regions with coral reefs.
5. **c) Halite, at about 10% of the original volume**
p. 143 — the order follows how concentrated the remaining water has become, so halite requires the most evaporation.
6. **b) records the direction of current or wind flow at the time of deposition**
pp. 145–146 — analyzed in detail it can also indicate the rate of flow and the amount of sediment available. Ripples help the same way, since their steepest face points downflow.

Part B · True or False

1 point each

7. **TRUE**
p. 144 — the cobbles inside a conglomerate had to exist before the conglomerate could form around them.
8. **FALSE**
Most fossils are found in sedimentary rock. Some occur in metamorphic rock, but fossils are rare in igneous rock because it forms from flowing molten material. (p. 147)

Part C · Short Answer

3 points each

9. **Clastic rocks are made of material transported as solid fragments (clasts); chemical rocks are made of material transported as ions dissolved in solution.**

Rubric: 1 pt clastic defined — composed mainly of material carried as solid fragments; examples include mudstone, shale, sandstone, conglomerate, breccia, and coal. 1 pt chemical defined — composed mainly of material carried as ions in solution (Na^+ , Ca^{2+} , HCO_3^-); examples include limestone, chert, banded iron formation, gypsum, and halite. 1 pt for one valid example of each. Bonus insight: the two overlap, since nearly all clastic rocks are cemented by minerals precipitated from dissolved ions. (pp. 135 and 141)

10. **They record the rise of free oxygen produced by photosynthetic bacteria, which converted dissolved iron in seawater into insoluble iron oxide that sank to the seafloor. They stopped forming when the oceans ran out of dissolved iron.**

Rubric: 1 pt — photosynthetic bacteria such as cyanobacteria evolved around 3500 Ma and consumed carbon dioxide to produce oxygen, but for about a billion years that oxygen was consumed by other processes; free oxygen began accumulating around 2400 Ma. 1 pt — over roughly 600 million years that oxygen converted soluble ferrous iron to insoluble ferric iron, forming hematite that settled onto the deep seafloor as BIF. 1 pt — by 1800 Ma little dissolved iron remained in the oceans, so BIF formation essentially stopped. (p. 143)